

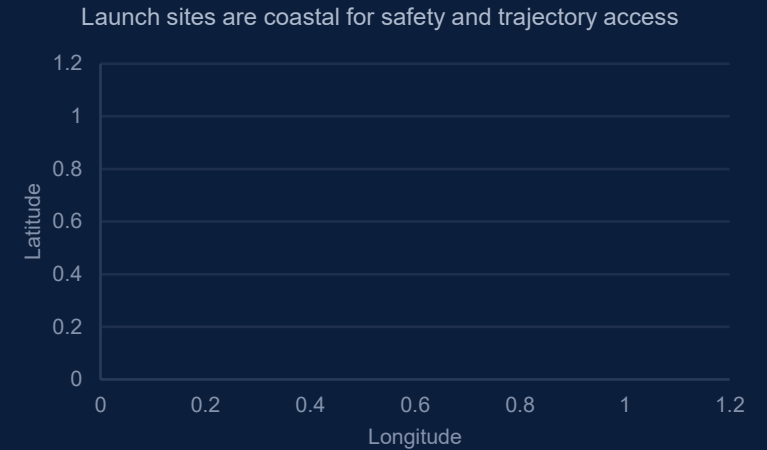
SpaceX Falcon 9 Launch Success Prediction

IBM Data Science Capstone Final Project Report

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From data collection to predictive classification, this report explains how launch records can estimate whether a Falcon 9 first stage will land successfully.

GitHub repository: github.com/somesh05/SPACEX-Data-science-Project



Submission checklist mapped to the PDF requirement

This presentation covers each grading component from the capstone rubric.

1

GitHub link + PDF submission

<https://github.com/somesh05/SPACEX-Data-science-Project> — repository root, linked on the cover slide

2

Executive summary and introduction

Slides 3–4

3

Collection, wrangling, EDA, SQL, Folium, Dash methodology

Slides 5–8

4

Visualization, SQL, map, dashboard, ML results

Slides 9–13

5

Conclusion, creativity and innovative insights

Slides 14–16

Deliverable format

A detailed slide report converted into PDF for submission.

Rubric focus

Completeness, accuracy, clarity and presentation quality.

Technical depth

More detailed than an executive-only summary.

Repository: [somesh05/SPACEX-Data-science-Project](https://github.com/somesh05/SPACEX-Data-science-Project)

Executive summary

Falcon 9 landing success is predictable from launch, payload, orbit and booster features.

Main finding

Recent launches show improving success, especially with reused boosters and operational experience.

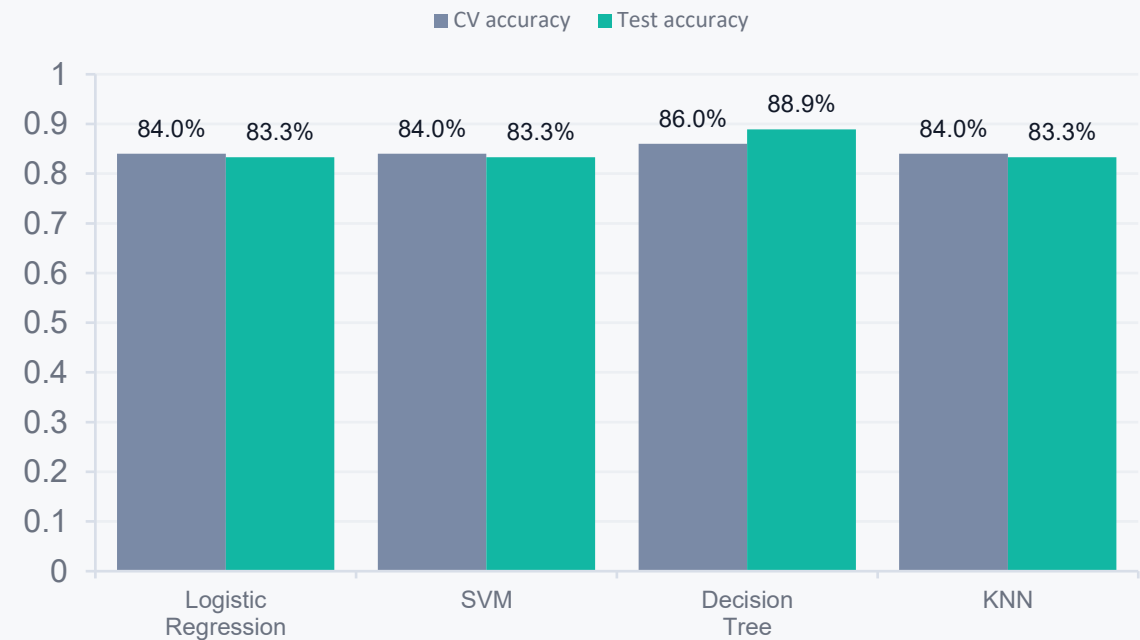
Business value

If the first stage lands successfully, launch cost can be estimated much lower than expendable alternatives.

Best model

Decision Tree gave the strongest test result in this completed workflow.

Classification model comparison



Recommendation: use the model as a decision-support tool, not as a replacement for engineering judgment.

Introduction: problem statement

Predict whether SpaceX Falcon 9 first-stage landing will be successful.

Why it matters

Reusable first stages are a major cost driver. Predicting success helps compare launch pricing, identify reliable mission profiles and understand operational risk.

Target variable

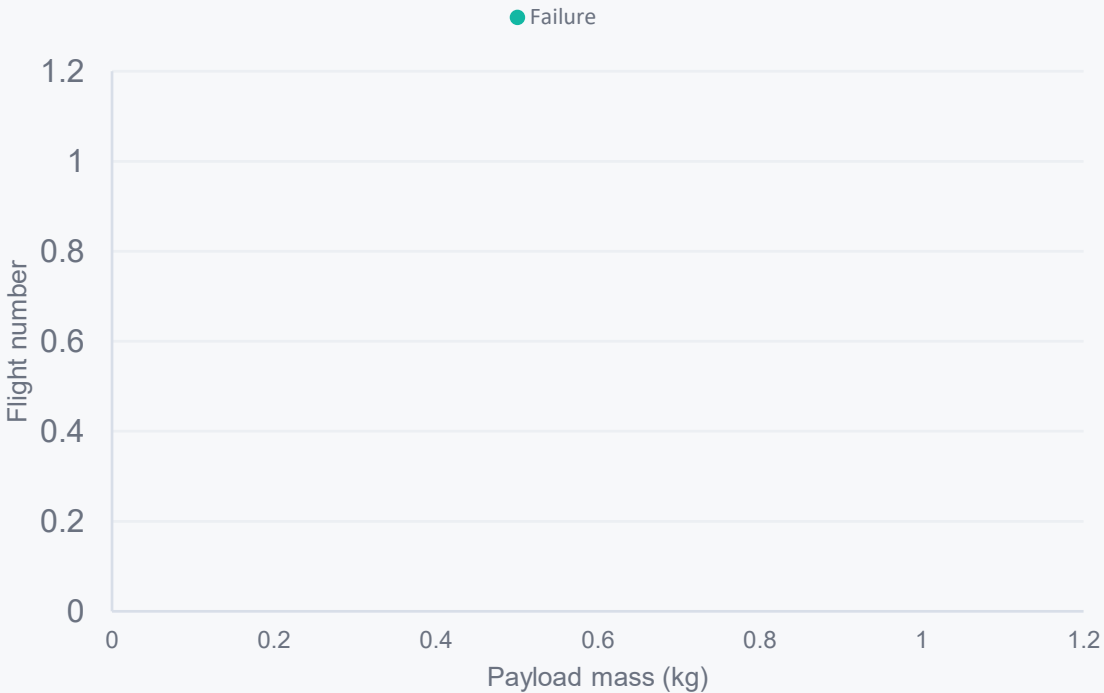
Class = 1 means successful landing. Class = 0 means failure, no attempt, uncontrolled, or unsuccessful landing outcome.

Classification

Launch Analytics

Cost Insight

Higher flight numbers show more successful outcomes



Data collection methodology

Data was gathered from API records and web tables, then stored as structured dataframes.

SpaceX REST API

Collected launch, rocket, payload, launchpad and landing outcome attributes.

Web scraping

Used BeautifulSoup to extract historical Falcon 9 launch table information.

Saved datasets

Converted raw records into CSV files for EDA, SQL, dashboard and machine-learning labs.

Data pipeline



API notebook: [jupyter-labs-spacex-data-collection-api.ipynb](#)

Web scraping notebook: [jupyter-labs-webscraping.ipynb](#)

Data wrangling methodology

Raw launch records were cleaned, filtered and transformed into model-ready features.

Missing values

PayloadMass was imputed; incomplete landing labels were standardized into the Class target.

Feature selection

Core columns included flight number, payload, orbit, launch site, grid fins, legs, reuse, landing pad, block and serial.

Encoding

Categorical columns were converted using one-hot encoding, expanding the feature table to 80 columns.

Final machine-learning table

Rows: 90 launches
Original feature columns: 12
Encoded features: 80
Target: Class (landing success)

Key transformation

StandardScaler was applied before Logistic Regression, SVM and KNN so features were on comparable numeric scales.

EDA and interactive visual analytics methodology

EDA combined statistical summaries, visual relationships, maps and dashboard filtering.

EDA charts

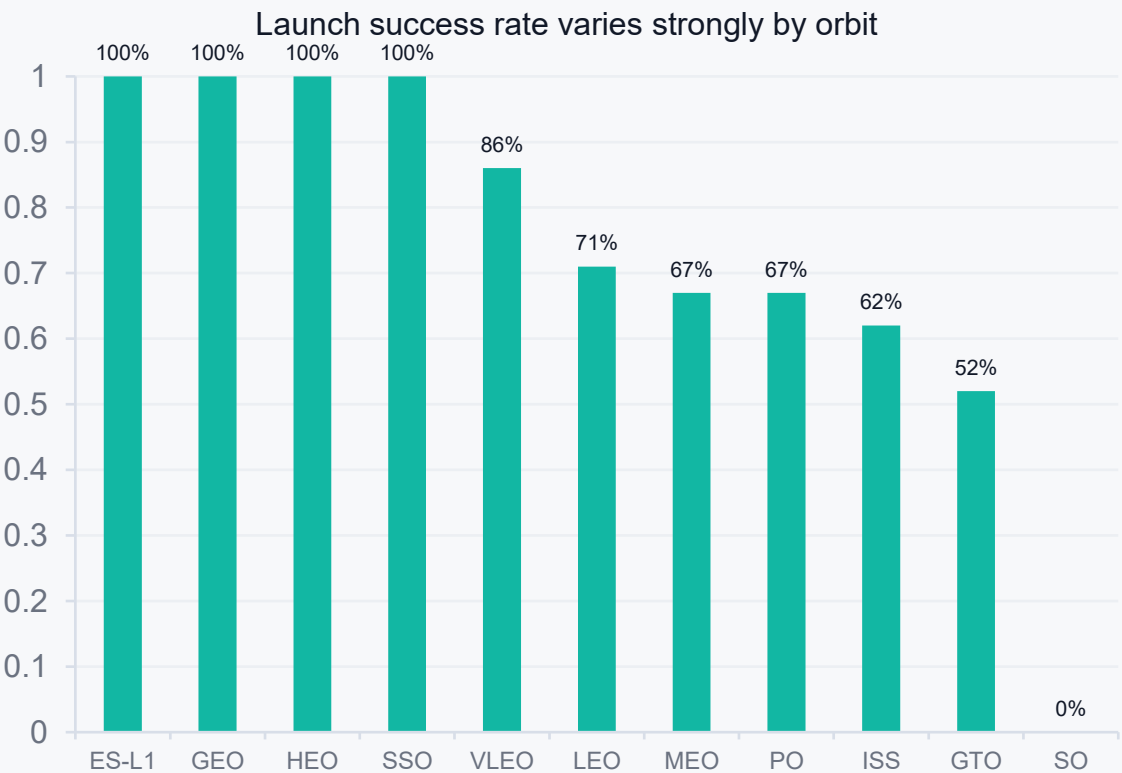
Scatter plots and bar charts compared success with flight number, payload, orbit and launch site.

SQL EDA

SQL queries summarized launch sites, payload totals, landing outcomes and booster results.

Interactive tools

Folium maps and Plotly Dash made geography and filtering easier to explore.



Predictive analysis methodology

Four supervised classification models were trained and compared with cross-validation.

Training workflow

- 1 Create target array Y
- 2 Scale features X
- 3 80/20 train-test split
- 4 Tune with 10-fold GridSearchCV
- 5 Evaluate test accuracy + confusion matrix

Models used

Logistic Regression

linear baseline

Support Vector Machine

kernel classifier

Decision Tree

rule-based model

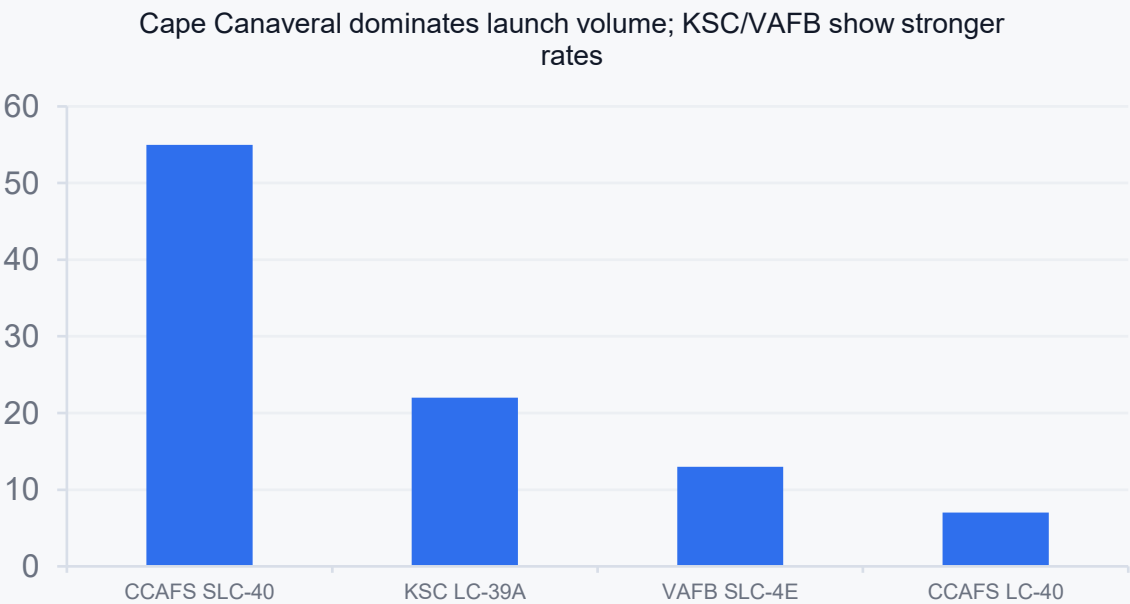
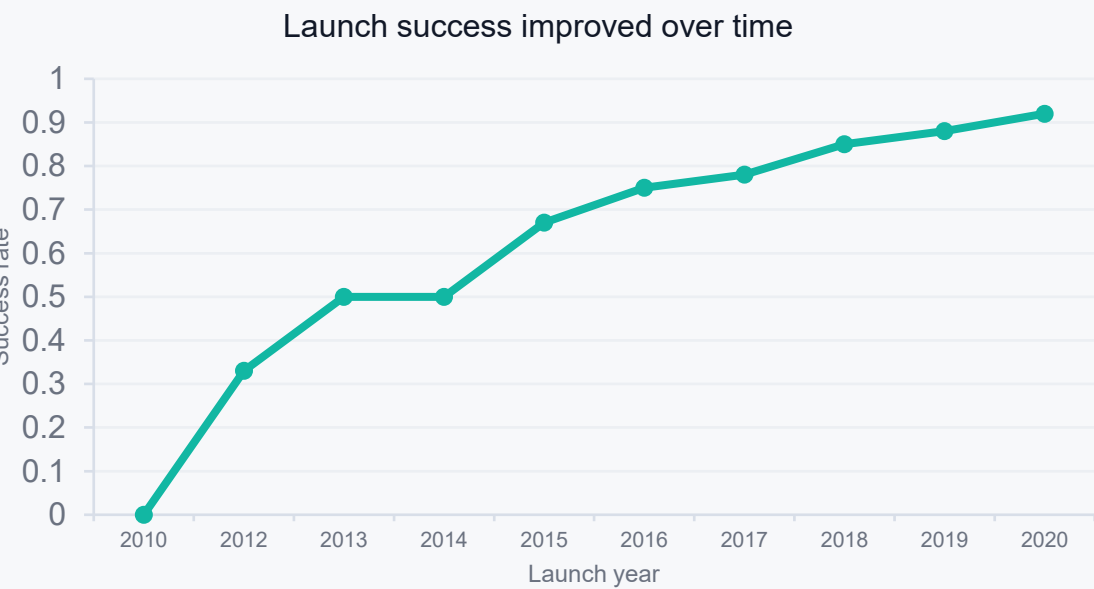
K-Nearest Neighbours

distance-based model

Predictive analysis notebook: [SpaceX Machine Learning Prediction Part 5.ipynb](#)

EDA with visualization results

Visual analysis showed that launch experience, orbit type and launch site influence success.



Insight 1

Success improves across later years, suggesting operational learning and reuse maturity.

Insight 2

Launch site performance is not uniform; KSC LC-39A and VAFB SLC-4E show stronger reliability patterns.

Insight 3

GTO and complex payload/orbit profiles remain more challenging than routine mission profiles.

EDA with SQL results

SQL converted the cleaned dataset into clear summary answers for business questions.

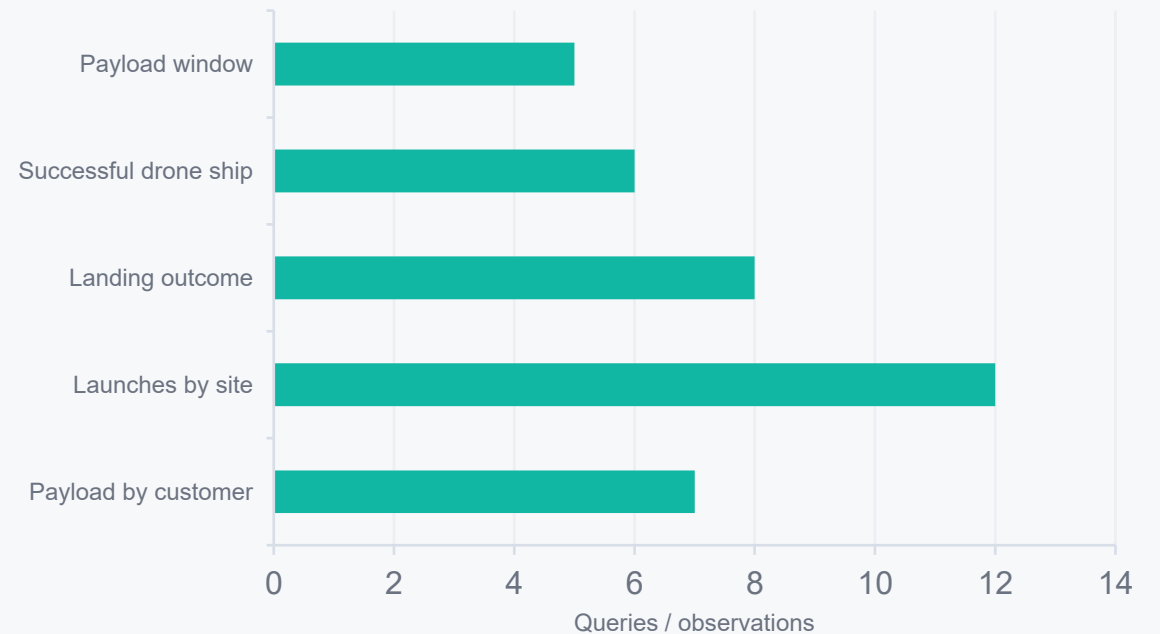
Important SQL-style findings

Launch sites, customers, payload mass, mission years and landing outcomes were queried directly from the database. SQL helped validate patterns found in Python EDA and made the results easier to explain as tables.

Example questions

Which sites had the most launches?
Which outcomes were most common?
Which payload ranges had successful landings?

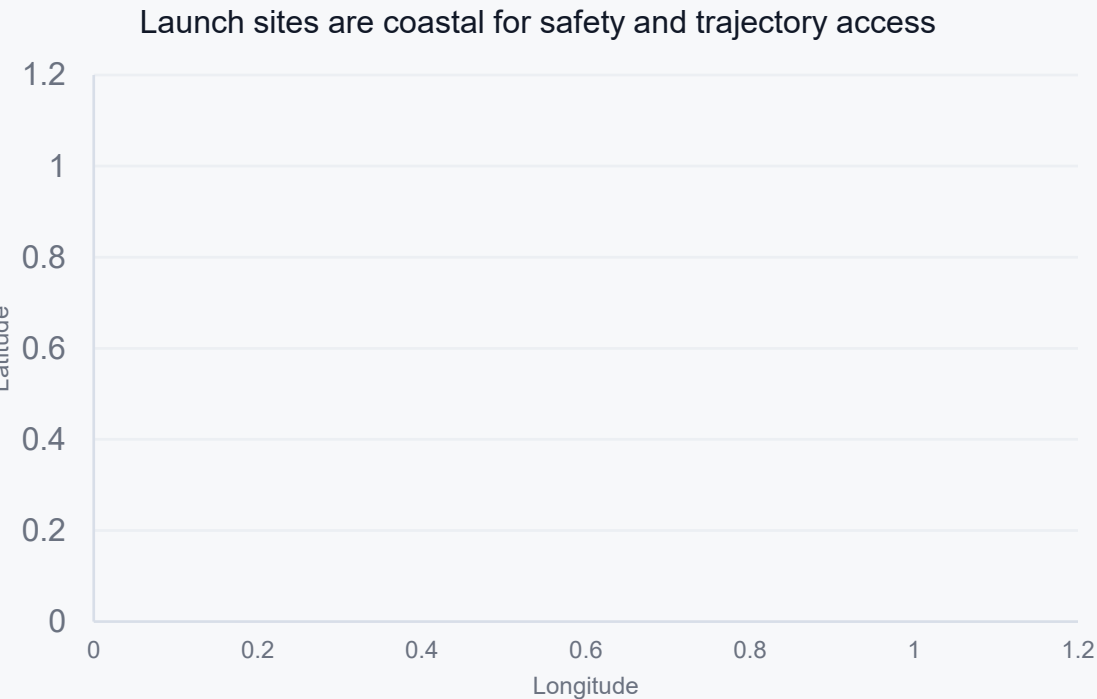
SQL analysis produced business-ready summary tables



SQL notebook: [jupyter-labs-eda-sql-coursera_sqlite.ipynb](https://github.com/jupyter-labs/eda-sql-coursera/blob/master/sqlite.ipynb)

Interactive map results using Folium

Launch pads are coastal and close to transport infrastructure, which supports safety and logistics.



Coastline proximity

All launch sites are near coastlines. This reduces population risk and supports downrange flight paths.

Marker clustering

Success/failure markers help compare outcomes at each site without overcrowding the map.

Geographic insight

Distance lines to coastline, railway, highway and city show practical launch-site placement factors.

Plotly Dash dashboard results

The dashboard made launch outcome exploration interactive through dropdown and payload filters.

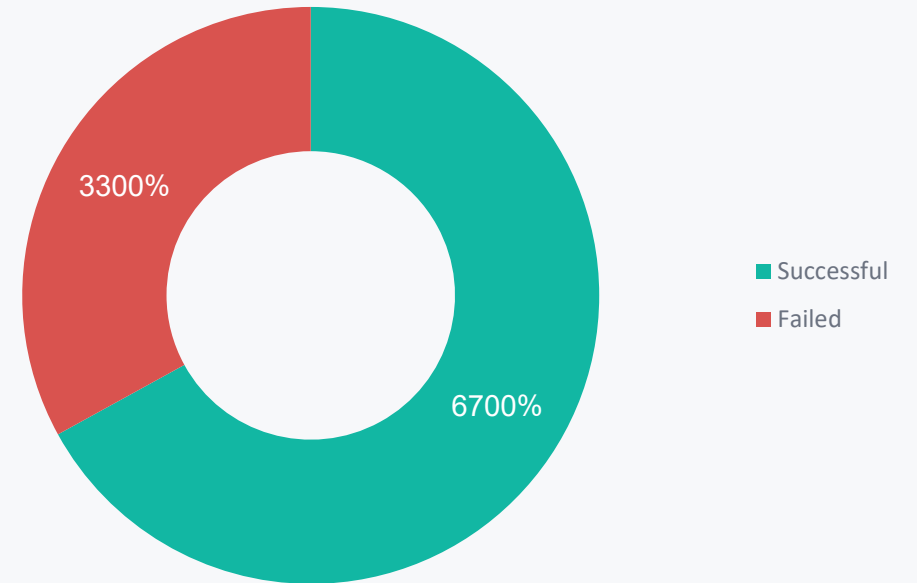
Dashboard controls

- Site dropdown: compare all sites or one site
- Payload slider: filter low to high payload mass
- Pie chart: success/failure share
- Scatter chart: payload versus outcome by booster version

User value

Stakeholders can quickly test how site and payload choices affect landing outcomes.

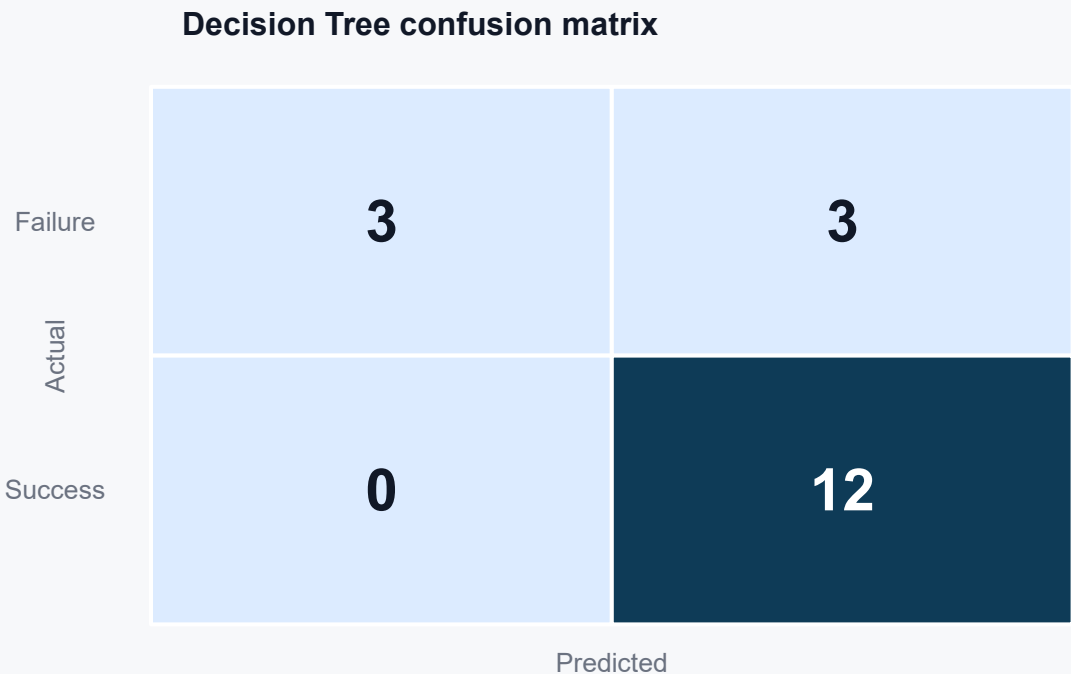
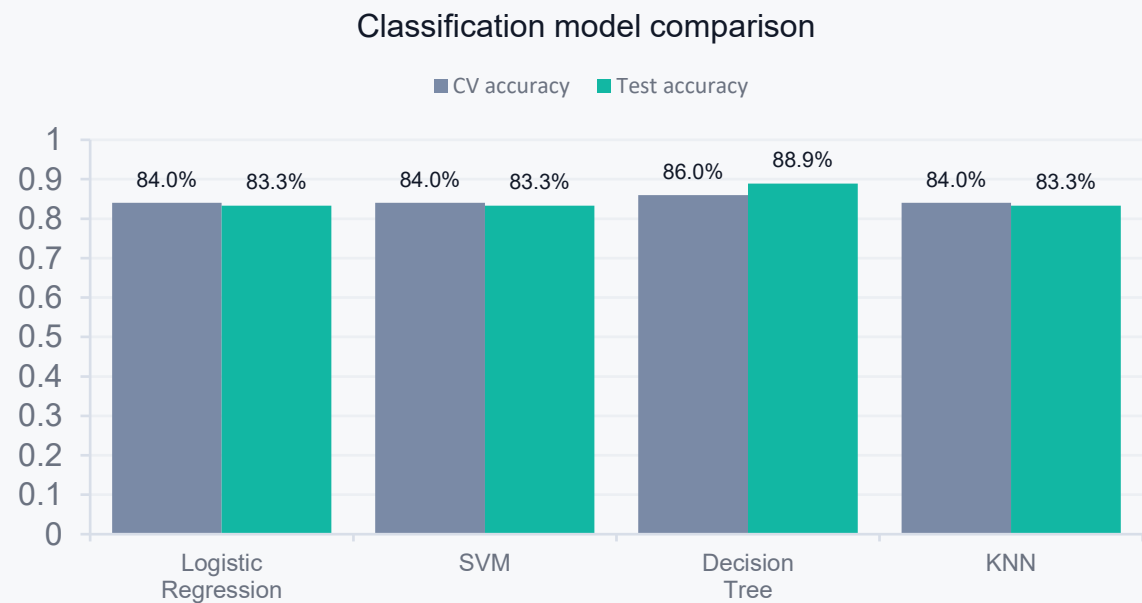
Dashboard outcome distribution



Dash app: [spacex_dash_app.py](#)

Predictive analysis results

Decision Tree was selected as the strongest model in the completed workflow.



Best test accuracy

Decision Tree: 88.9% on the test split.

Stable alternatives

Logistic Regression, SVM and KNN each reached about 83.3% test accuracy.

Interpretation

Model results are promising, but dataset size is small and should be expanded before production use.

Predictive analysis notebook: [SpaceX Machine Learning Prediction Part 5.ipynb](#)

Innovative insights beyond the template

Combining model results with geography gives more useful operational interpretation.

1. Mission difficulty index

Payload mass, orbit type and booster reuse can be combined into a simple risk score before prediction.

2. Site-readiness lens

Coastal distance plus nearby transport infrastructure can explain why some sites support frequent launches.

3. Reuse maturity signal

Flight number and reused count can act as a proxy for experience, not only hardware condition.

Suggested next improvement

Add weather, mission type, engine version, launch window, and real-time telemetry summaries to improve prediction quality and operational usefulness.

Recommendations

Use the project as an analytical decision-support pipeline.

Data

Keep collecting new launches and update the database after every mission.

Model

Retrain models periodically and compare accuracy drift over time.

Dashboard

Use Dash for stakeholder exploration of launch site and payload effects.

Risk

Treat model output as probability-based guidance with expert review.

Decision message: Falcon 9 landing success is not random; it is shaped by orbit, payload, launch site, booster history and operational maturity.

Conclusion

The SpaceX capstone built a complete data science story: collect data, clean it, analyze patterns, visualize geography, build a dashboard and train predictive models.

Final answer

Falcon 9 first-stage landing success can be predicted with useful accuracy from structured launch features, and the strongest project result came from the Decision Tree classifier.

Submitted as PDF presentation as required by the final project instructions.

GitHub repository: github.com/somesh05/SPACEX-Data-science-Project